

MANGAMMAGARI GNANA CHAITHANYA

Deep Learning Engineer | Agentic AI & GenAI Specialist

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Deep Learning Engineer with 2+ years building production LLM systems, multi-agent pipelines, RAG architectures, and predictive ML. Delivered end-to-end AI systems across education, healthcare, manufacturing, and enterprise automation. Hands-on with Gemini 2.5 Pro, GPT-4o, LangChain, LangGraph, CrewAI, FastAPI, Celery, and cloud deployments on GCP, AWS, and Azure.

PROFESSIONAL EXPERIENCE

Deep Learning Engineer · [DataToBiz](#)

Jan 2026 – Present

- Automated Grading System — Built a 5-stage async pipeline (Flask + Celery + Redis) processing handwritten exam scripts end-to-end: paper parsing via Gemini 2.5 Pro Vision → answer extraction → LLM evaluation against marking memo → score calculation → annotated PDF generation with red-pen marks uploaded to AWS S3. Handles 1,000+ scripts/cycle; reduced turnaround from days to minutes, improved marking consistency by 92% across 109 automated test cases.
- AI Persona Emulation — Engineered a 6-stage async LLM pipeline (FastAPI + Celery + Redis) generating synthetic brand personas from audience demographic data for a global media agency. Built multi-provider LLM abstraction (GPT-4o, Claude, Gemini, Perplexity) with automatic failover. PostgreSQL + Azure Blob, deployed via Docker. Streamlined persona workflows by 70%.
- Voice + Chat Bot — Dual voice/text interface with sub-200ms latency via WebRTC + Retell AI; enterprise auth, deployed on Hetzner cloud serving real-time concurrent sessions.

AI Engineer Intern · [DataToBiz](#)

Jul 2025 – Dec 2025

- BMR Predictive Maintenance — Built a 9-stage ML pipeline (BiLSTM + Autoencoder) for early prediction of medical device failures: ingestion → cleaning → feature engineering → SMOTE → BiLSTM training → SHAP explainability → FastAPI serving. Reduced unplanned failures by 15% and accelerated root-cause diagnosis by 40% via SHAP dashboards. Served via FastAPI + Streamlit on PostgreSQL.
- AI Pitch Deck Generator — 3 specialised CrewAI agents (Research via SerperDev, Knowledge via Pinecone RAG, Content via GPT-4o-mini) with Streamlit UI + FastAPI REST endpoint. Auto-exports editable .pptx. Cut pitch preparation from days to under 30 minutes.
- Sales Intelligence Tool (AI/LLM Layer) — Built GPT-4o synthesis engine aggregating 12+ sources (SearXNG, Google News, LinkedIn, Meta Ad Library, Nielsen) into 11-tab company briefs via SSE streaming. Responsible for prompt engineering, retry logic, and structured output validation.

Data Science Intern · [AISPRY](#)

Mar 2024 – Mar 2025

- Medical Imaging — Improved dermatology triage accuracy by 35% using YOLOv8 for skin lesion detection; improved IVF embryo grading accuracy by 25% with a custom CNN classifier.
- Legal Doc Processing & Forecasting — Built an LLM clause-extraction pipeline that cut contract review time by 40%; trained an LSTM coal-demand model delivering ~Rs. 1L in annual savings via real-time Streamlit dashboards.
- Emotion Detection Research — Benchmarked DeepFace, FER, and RMN models on automotive driver-monitoring video datasets; produced per-frame emotion CSVs, confidence scores, and annotated output videos for engineering evaluation.

PERSONAL PROJECTS

NetworkSecurity — Phishing Detection with Full MLOps Stack | [GitHub](#) | Python · MLflow · Docker · GitHub Actions CI/CD · AWS ECR / EC2 · FastAPI

- Built phishing URL detection as a fully operationalised ML system — not a notebook. Implemented modular training pipeline with MLflow experiment tracking and model registry, containerised the inference service with Docker, and wired GitHub Actions to auto-build and push to AWS ECR on every merge to main. FastAPI prediction endpoint deployed on EC2. Chose this problem specifically to practice the operational gap most ML projects skip: reproducible retraining, versioned models, and zero-downtime deployment.

AdvRAG — Hybrid Vector + Keyword Retrieval System | [GitHub](#) | Python · LangChain · FAISS · ChromaDB · BM25 · OpenAI / Gemini Embeddings

- Investigated why vector-only RAG fails on keyword-dense queries (product codes, legal clauses, IDs) and built a hybrid retriever fusing FAISS semantic search with BM25 keyword scoring via reciprocal rank fusion. Systematically compared recursive, semantic, and fixed chunking strategies across Hit Rate and MRR metrics, documenting the

latency vs. accuracy trade-off at each configuration — the kind of retrieval audit production RAG systems require before going live.

TravelAgent — Graph-Based Multi-Agent Travel Planner | [GitHub](#) | *Python · Gemini Pro · LangGraph · LangChain · Flask · Bootstrap*

- Designed a LangGraph directed state graph that decomposes an open-ended travel request into four specialised agents — Research, Budgeting, Routing, and Reporting — each reading and writing to a shared state object. Chose graph-based orchestration over a linear chain to allow conditional branching (e.g. budget constraints re-routing the research agent). Outputs a structured JSON/Markdown itinerary served through a Flask UI.

MediConsult AI — Dual-Modality Medical Assistant | [GitHub](#) | *Python · LangChain · Groq LLaMA3 · FAISS · HuggingFace MiniLM · YOLOv8 · Streamlit*

- Solved two distinct medical assistance problems in one app: semantic Q&A over the Gale Medical Encyclopedia using FAISS vector store + MiniLM embeddings + Groq LLaMA3 (chosen for low-latency inference over GPT-4 for this use case), and visual skin condition screening via a YOLOv8 detection model on uploaded images. Multilingual support via translation layer. Deployed on Vercel with Render cloud config.

TalentScout — Automated AI Recruitment Screener | [GitHub](#) | *Python · LangChain · Gemini · Pydantic · Streamlit*

- Automated the part of hiring that costs recruiters the most time: initial screening. Parses uploaded resumes, scores candidate-to-job-description fit on a 0–10 scale using Gemini, generates role-specific technical interview questions per candidate profile, and produces a sentiment-analysed pass/fail decision — all in one pipeline. Structured outputs enforced via Pydantic throughout to eliminate hallucinated field values.

TECHNICAL SKILLS

LLMs & GenAI: GPT-4/4o, Gemini 2.5 Pro/Vision, LLaMA 3, Mistral, Claude API, QLoRA, RAG, Prompt Engineering, Structured Output (Pydantic / Instructor)

Agent Frameworks: LangChain, LangGraph, AutoGen, CrewAI — production & project deployments

Backend & Async: FastAPI, Flask, Celery, Redis, WebRTC, Retell AI, SSE streaming, n8n

Deep Learning: TensorFlow, PyTorch, CNN, YOLOv8, BiLSTM, Autoencoders, SHAP, LIME

Cloud & MLOps: GCP (Document AI, Vision AI), AWS (S3, EC2, ECR), Azure Blob, Docker, MLflow, GitHub Actions CI/CD, Airflow

Databases: PostgreSQL, MySQL, MongoDB, Pinecone, ChromaDB, FAISS, Milvus, SQLite

Languages: Python (advanced), SQL, Node.js / Express

CERTIFICATIONS

- AI & Deep Learning** — 360DigiTMG
- Data Science** — 360DigiTMG
- Python for Data Science** — 360DigiTMG
- SQL for Data Analysis** — 360DigiTMG
- Power BI** — 360DigiTMG

EDUCATION

B.Tech — Computer Science (AI & ML Specialization)

CGPA: 8.17 / 10

Siddharth Institute of Engineering and Technology | 2020 – 2024

Relevant coursework: Deep Learning, NLP, Computer Vision, Machine Learning, DSA